Supplementary Material for "A Bayesian Critique of a Rank-Based Method for Surrogate Marker Evaluation" by Carlotti and Parast

Web Appendix A. Derivation of θ and δ as Expectations

This appendix derives expressions for θ and δ as expectations over the covariate distribution.

Web Appendix A.1 *Derivation of θ as an Expectation over the Covariate Distribution*

Under the model in Equation (3), it is easy to see that

$$Y_{1i} - Y_{0i} \mid X_i = x \overset{\text{ind.}}{\sim} \mathcal{N}\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right), \quad i = 1, \ldots, n, \ \forall\, x \in \mathcal{X}.$$

Therefore, we obtain this analytical expression for V^x_Y:

$$\begin{aligned}
V^x_Y &= \mathbb{P}\left(Y_{1i} > Y_{0i} \mid X_i = x\right) \\
&= \mathbb{P}\left(Y_{1i} - Y_{0i} > 0 \mid X_i = x\right) \\
&= \mathbb{P}\left(\frac{Y_{1i} - Y_{0i} - \mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}} > \frac{-\mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}} \mid X_i = x\right) \\
&= \mathbb{P}\left(W > \frac{-\mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) \\
&= 1 - \Phi\left(\frac{-\mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) \\
&= \Phi\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right), \ \forall\, x \in \mathcal{X},
\end{aligned}$$

where $W \sim \mathcal{N}(0,1)$ and Φ is the standard normal CDF. Then, by the law of total probability, we can derive an analytical expression for V_Y as follows:

$$\begin{aligned}
V_Y &= \mathbb{P}\left(Y_{1i} > Y_{0i}\right) \\
&= \int_{\mathcal{X}} V^x_Y \, f(x) \, \mathrm{d}x \\
&= \int_{\mathcal{X}} \Phi\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) f(x) \, \mathrm{d}x.
\end{aligned}$$

Applying a similar logic to V_S, we obtain this analytical expression for θ:

$$\theta = \int_{\mathcal{X}} \theta^x \, f(x) \, \mathrm{d}x,$$

where we denote

$$\theta^x = \Phi\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) - \Phi\left(\frac{\mu^x_{S_1} - \mu^x_{S_0}}{\sqrt{\Sigma^x_{S_1S_1} + \Sigma^x_{S_0S_0} - 2\Sigma^x_{S_1S_0}}}\right), \ \forall\, x \in \mathcal{X}.$$

Web Appendix A.2 *Derivation of δ as an Expectation over the Covariate Distribution*

Under the model in Equation (3), it is easy to see that

$$Y_{1i} - Y_{0j} \mid X_i = x_i, X_j = x_j \sim \mathcal{N}\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right), \ \forall\, x_i, x_j \in \mathcal{X},\ i, j = 1, \ldots, n.$$

Therefore, we obtain this analytical expression for $U_Y^{x_ix_j}$:

$$\begin{aligned}
U_Y^{x_ix_j} &= \mathbb{P}\left(Y_{1i} > Y_{0j} \mid X_i = x_i, X_j = x_j\right) \\
&= \mathbb{P}\left(\frac{Y_{1i} - Y_{0j} - \mu_{Y_1}^{x_i} + \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}} > \frac{-\mu_{Y_1}^{x_i} + \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}} \mid X_i = x_i, X_j = x_j\right) \\
&= \mathbb{P}\left(W > \frac{-\mu_{Y_1}^{x_i} + \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right) \\
&= 1 - \Phi\left(\frac{-\mu_{Y_1}^{x_i} + \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right) \\
&= \Phi\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right), \ \forall\, x_i, x_j \in \mathcal{X},\ i, j = 1, \ldots, n.
\end{aligned}$$

Then, by the law of total probability, we can derive an analytical expression for U_Y as follows:

$$\begin{aligned}
U_Y &= \mathbb{P}\left(Y_{1i} > Y_{0j}\right) \\
&= \int_{\mathcal{X}}\int_{\mathcal{X}} U_Y^{x_i,x_j}\, f(x_i)\, f(x_j)\, \mathrm{d}x_i\, \mathrm{d}x_j \\
&= \int_{\mathcal{X}}\int_{\mathcal{X}} \Phi\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right) f(x_i)\, f(x_j)\, \mathrm{d}x_i\, \mathrm{d}x_j.
\end{aligned}$$

Applying a similar logic to U_S, we obtain this analytical expression for δ:

$$\delta = \int_{\mathcal{X}}\int_{\mathcal{X}} \delta^{x_ix_j}\, f(x_i)\, f(x_j)\, \mathrm{d}x_i\, \mathrm{d}x_j,$$

where we denote

$$\delta^{x_ix_j} = \Phi\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right) - \Phi\left(\frac{\mu_{S_1}^{x_i} - \mu_{S_0}^{x_j}}{\sqrt{\Sigma_{S_1S_1}^{x_i} + \Sigma_{S_0S_0}^{x_j}}}\right), \ \forall\, x_i, x_j \in \mathcal{X}.$$

Web Appendix A.3 *Derivation in the Condionally Orthogonal and Homoscedastic Setting*

Under the assumption of conditional orthogonality stated in the paper, it follows that

$$V_Y^x = U_Y^{xx} \text{ and } V_S^x = U_S^{xx},\ \forall\, x \in \mathcal{X},$$

so that

$$\theta^x = \delta^{xx},\ \forall\, x \in \mathcal{X}. \tag{A.1}$$

Under the assumption of conditional homoscedasticity stated in the paper, it follows that

$$\theta = \mathbb{E}_{X_i \sim f}[g(X_i, X_i)]\,, \quad \delta = \mathbb{E}_{(X_i, X_j) \sim f^2}[g(X_i, X_j)]\,,$$

where we denote

$$g(x_i, x_j) = \Phi\left(\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}\right) - \Phi\left(\mu_{S_1}^{x_i} - \mu_{S_0}^{x_j}\right),\ \forall\, x_i, x_j \in \mathcal{X}. \tag{A.2}$$

Web Appendix B. Derivation of $|\theta - \delta|$ in the Binary Covariate Setting

Under the setting described in Section 2.4.1 of the main paper, θ and δ reduce to

$$\theta = \frac{\theta^0 + \theta^1}{2}, \quad \delta = \frac{\delta^{00} + \delta^{11} + \delta^{01} + \delta^{10}}{4}.$$

Based on Equation (A.1), we have that

$$|\theta - \delta| = \frac{|\delta^{00} + \delta^{11} - \delta^{01} - \delta^{10}|}{4}. \tag{A.3}$$

This expression highlights that the distance between the two parameters is driven by differences between comparisons within the same covariate group and comparisons across different covariate groups. Importantly, this effect does not arise from treatment effect heterogeneity.

To derive a formula for $|\theta - \delta|$, let us further simplify the setting by assuming that S is a perfect surrogate for Y with homogeneous treatment effects across covariate groups:

$$\mu_{Y_1}^x - \mu_{Y_0}^x = \mu_{S_1}^x - \mu_{S_0}^x = \Delta > 0,\ \forall\, x \in \{0, 1\}.$$

In this setting, we can easily compute the terms in Equation (A.3) using Equation (A.2):

$$\delta^{00} = \Phi(\Delta) - \Phi(\Delta) = 0,$$

$$\delta^{11} = \Phi(\Delta) - \Phi(\Delta) = 0,$$

$$\delta^{01} = \Phi\big(\Delta + \mu^0_{Y_0} - \mu^1_{Y_0}\big) - \Phi\big(\Delta + \mu^0_{S_0} - \mu^1_{S_0}\big)\,,$$

$$\delta^{10} = \Phi\big(\Delta + \mu^1_{Y_0} - \mu^0_{Y_0}\big) - \Phi\big(\Delta + \mu^1_{S_0} - \mu^0_{S_0}\big)\,.$$

Hence, we conclude that

$$|\theta - \delta| = \frac{1}{4}\left|\Phi(\Delta + d_Y) - \Phi(\Delta + d_S) + \Phi(\Delta - d_Y) - \Phi(\Delta - d_S)\right|,$$

where we denote $d_Y = \mu^0_{Y_0} - \mu^1_{Y_0}$ and $d_S = \mu^0_{S_0} - \mu^1_{S_0}$.

Web Appendix B.1 *Supremum of* $|\theta - \delta|$

We now compute analytically the supremum of $|\theta - \delta|$ with respect to d_Y and d_S. First, observe that the distance $|\theta - \delta|$ can be rewritten as

$$|\theta - \delta| = \frac{1}{4}\left|g_\Delta(d_Y) - g_\Delta(d_S)\right|,$$

where

$$g_\Delta(x) = \Phi(\Delta + x) + \Phi(\Delta - x),\ \forall\, x \in \mathbb{R}.$$

Then, it follows that

$$\sup_{d_Y, d_S}\{|\theta - \delta|\} = \frac{1}{4}\left(\sup_x\{g_\Delta(x)\} - \inf_x\{g_\Delta(x)\}\right)$$

Moreover, since

$$g_\Delta(-x) = g_\Delta(x),\ \forall\, x \in \mathbb{R},$$

we only have to optimize g_Δ over $\mathbb{R}_+$. Observe that the derivative of g_Δ is given by

$$\begin{aligned} g'_\Delta(x) &= \phi(\Delta + x) - \phi(\Delta - x) \\ &= \frac{1}{\sqrt{2\pi}}\left(e^{-\frac{(\Delta+x)^2}{2}} - e^{-\frac{(\Delta-x)^2}{2}}\right),\ \forall\, x \in \mathbb{R}_+, \end{aligned}$$

where ϕ is the standard normal PDF. Then, the critical points of g_Δ satisfy this equation:

$$\phi(\Delta + x) = \phi(\Delta - x) \iff (\Delta + x)^2 = (\Delta - x)^2,$$

which has only one solution $x = 0$, since we assumed that $\Delta > 0$. Moreover, we have that

$$g'_\Delta(x) < 0, \ \forall\, x \in \mathbb{R}_+.$$

This implies that g_Δ is strictly decreasing over $\mathbb{R}_+$, so that

$$\sup_x \{g_\Delta(x)\} = g_\Delta(0) = 2\Phi(\Delta),$$

$$\inf_x \{g_\Delta(x)\} = \lim_{x \to +\infty} g_\Delta(x) = 1.$$

Hence, we conclude that

$$\sup_{d_Y, d_S} \{|\theta - \delta|\} = \frac{1}{4}\left(2\Phi(\Delta) - 1\right) = \frac{1}{2}\Phi(\Delta) - \frac{1}{4}.$$

Web Appendix C. Derivation of θ and δ in the Gaussian Covariate Setting

Using the setting from Section 2.4.2, we express θ and δ via Equation (A.2) as follows:

$$\theta = \mathbb{E}_{X_i \sim f}[\Phi\left(X_i(\beta_{Y_1} - \beta_{Y_0})\right) - \Phi\left(X_i(\beta_{S_1} - \beta_{S_0})\right)],$$

$$\delta = \mathbb{E}_{(X_i, X_j) \sim f^2}[\Phi\left(X_i\beta_{Y_1} - X_j\beta_{Y_0}\right) - \Phi\left(X_i\beta_{S_1} - X_j\beta_{S_0}\right)].$$

Web Appendix C.1 *Derivation of θ*

Using the perfect surrogacy assumption in Equation (4), we have that

$$\begin{aligned}\theta &= \mathbb{E}_{X_i \sim f}[\Phi\left(X_i(\beta_{Y_1} - \beta_{Y_0})\right) - \Phi\left(X_i(\beta_{S_1} - \beta_{S_0})\right)] \\ &= \mathbb{E}_{X_i \sim f}[\Phi\left(X_i\Delta\right) - \Phi\left(X_i\Delta\right)] = 0.\end{aligned}$$

Web Appendix C.2 *Derivation of δ*

Observe that by the Gaussian covariate assumption, we have that

$$X_i\beta_{Y_1} - X_j\beta_{Y_0} \sim \mathcal{N}\left(m(\beta_{Y_1} - \beta_{Y_0}), v^2(\beta_{Y_1}^2 + \beta_{Y_0}^2)\right), \quad i, j = 1, \ldots, n,$$

$$X_i\beta_{S_1} - X_j\beta_{S_0} \sim \mathcal{N}\left(m(\beta_{S_1} - \beta_{S_0}), v^2(\beta_{S_1}^2 + \beta_{S_0}^2)\right), \quad i, j = 1, \ldots, n.$$

Therefore, we can derive an expanded analytical expression for δ as follows:

$$\delta = \mathbb{E}_{(X_i, X_j) \sim f^2}[\Phi\left(X_i\beta_{Y_1} - X_j\beta_{Y_0}\right)] - \mathbb{E}_{(X_i, X_j) \sim f^2}[\Phi\left(X_i\beta_{S_1} - X_j\beta_{S_0}\right)]$$

$$= \Phi\left(\frac{m[\beta_{Y_1} - \beta_{Y_0}]}{\sqrt{1 + v^2[\beta_{Y_1}^2 + \beta_{Y_0}^2]}}\right) - \Phi\left(\frac{m[\beta_{S_1} - \beta_{S_0}]}{\sqrt{1 + v^2[\beta_{S_1}^2 + \beta_{S_0}^2]}}\right),$$

where the last line follows from the following property of Gaussian random variables.

LEMMA 1: *Let $W \sim \mathcal{N}(m, v^2)$ for some $m \in \mathbb{R}$ and $v \in \mathbb{R}_+$. Then, we have that*

$$\mathbb{E}_{W\sim\mathcal{N}(m,v^2)}[\Phi(W)] = \Phi\left(\frac{m}{\sqrt{1+v^2}}\right).$$

Web Appendix C.3 *Supremum of $|\theta - \delta|$*

We now compute analytically the supremum of $|\theta - \delta|$ with respect to $\boldsymbol{\beta}$. First, observe that under the perfect surrogacy assumption in Equation (4), we have that

$$\begin{aligned}
|\theta - \delta| &= |\delta| \\
&= \left|\Phi\left(\frac{m\Delta}{\sqrt{1 + v^2[(\beta_{Y_0} + \Delta)^2 + \beta_{Y_0}^2]}}\right) - \Phi\left(\frac{m\Delta}{\sqrt{1 + v^2[(\beta_{S_0} + \Delta)^2 + \beta_{S_0}^2]}}\right)\right| \\
&= |h_{\Delta,m,v}(\beta_{Y_0}) - h_{\Delta,m,v}(\beta_{S_0})|,
\end{aligned}$$

where

$$h_{\Delta,m,v}(x) = \Phi\left(\frac{m\Delta}{\sqrt{1 + v^2[(x + \Delta)^2 + x^2]}}\right), \ \forall\, x \in \mathbb{R}.$$

Then, it follows that

$$\sup_{\boldsymbol{\beta}}\{|\theta - \delta|\} = \sup_x\{h_{\Delta,m,v}(x)\} - \inf_x\{h_{\Delta,m,v}(x)\}.$$

Notice that since x only appears in the denominator of the argument of Φ, finding the extrema of $h_{\Delta,m,v}$ is equivalent to finding the extrema of the function $k_{\Delta,m,v}$ defined as follows:

$$k_{\Delta,m,v}(x) = 1 + v^2[(x + \Delta)^2 + x^2] = 1 + v^2(2x^2 + 2\Delta x + \Delta^2), \ \forall\, x \in \mathbb{R}.$$

It is easy to see that

$$\begin{aligned}
\sup_x\{k_{\Delta,m,v}(x)\} &= \lim_{x\to+\infty} k_{\Delta,m,v}(x) = +\infty, \\
\inf_x\{k_{\Delta,m,v}(x)\} &= k_{\Delta,m,v}\left(-\frac{\Delta}{2}\right) = 1 + \frac{v^2\Delta^2}{2},
\end{aligned}$$

where we omit the algebraic details for brevity. Therefore, we have that

$$\sup_{\boldsymbol{\beta}}\{|\theta-\delta|\} = \left|\Phi\left(\frac{m\Delta}{\sqrt{1+\frac{v^2\Delta^2}{2}}}\right) - \Phi\left(0\right)\right|$$
$$= \left|\Phi\left(\frac{m}{\sqrt{\frac{1}{\Delta^2}+\frac{v^2}{2}}}\right) - \frac{1}{2}\right|,$$

which is strictly increasing in $|\Delta|$, so that

$$\sup_{\boldsymbol{\beta},\Delta}\{|\theta-\delta|\} = \lim_{|\Delta|\to+\infty}\left|\Phi\left(\frac{m}{\sqrt{\frac{1}{\Delta^2}+\frac{v^2}{2}}}\right) - \frac{1}{2}\right|$$
$$= \left|\Phi\left(\frac{m\sqrt{2}}{v}\right) - \frac{1}{2}\right|.$$

Web Appendix D. Regularity and Posterior Consistency

This appendix provides a formal definition of regularity for Bayesian models and states Schwartz's theorem, which is a fundamental result on posterior consistency of regular models.

Definition 1 (Regularity of a Bayesian model): Let a Bayesian model be defined as

$$D_i \mid \phi \overset{\text{i.i.d.}}{\sim} f_\phi, \quad i = 1, \ldots, n,$$
$$\phi \sim \pi,$$

where D_i is the observed data for unit i, f_ϕ is the likelihood function parametrized by ϕ, and π is a prior distribution over the parameter ϕ. We say that the Bayesian model is *regular* if the following conditions hold:

- **Kullback-Leibler support of the prior**: The prior π assigns positive mass to every Kullback-Leibler neighborhood of the true parameter ϕ^*:

  $$\pi\left(\{\phi : D_{\mathrm{KL}}(f_{\phi^*}\|f_\phi) < \varepsilon\}\right) > 0,\ \forall\, \varepsilon \in \mathbb{R}_+,$$

  where $D_{\mathrm{KL}}(f_{\phi^*}\|f_\phi)$ is the Kullback-Leibler divergence of f_ϕ from f_{ϕ^*}.

- **Identifiability**: The model is identifiable:

  $$f_\phi \neq f_{\phi'},\ \forall\, \phi \neq \phi'.$$

THEOREM 1 (Schwartz's theorem (Schwartz, 1965)): *Let the data be generated from the following process:*

$$D_i \overset{i.i.d.}{\sim} f_{\phi^*}, \quad i = 1, \ldots, n,$$

and let the Bayesian model be defined as

$$D_i \mid \phi \overset{i.i.d.}{\sim} f_{\phi}, \quad i = 1, \ldots, n,$$

$$\phi \sim \pi.$$

If the model is regular according to Definition 1, then we have that

$$\pi\left(U^c \mid D_{1:n}\right) \overset{P_{\phi^*}}{\to} 0,$$

for every neighborhood U of ϕ^*, where P_{ϕ^*} is the probability measure induced by f_{ϕ^*}.

Web Appendix E. Derivation of the Bayes Factor for the Validation Threshold

Under the model described in Section 3.3 of the main paper, the marginal likelihoods under the null and alternative hypotheses can be explicitly stated as follows:

$$\begin{aligned}
L\left(\widehat{V}_S \mid H_0\right) &= \binom{n}{n\,\widehat{V}_S} \left(\frac{1}{2}\right)^{n\,\widehat{V}_S} \left(1 - \frac{1}{2}\right)^{n - n\,\widehat{V}_S} \\
&= \binom{n}{n\,\widehat{V}_S} \frac{1}{2^n}, \\
L\left(\widehat{V}_S \mid H_1\right) &= \int_{\frac{1}{2}}^{1} \binom{n}{n\,\widehat{V}_S} v_S^{n\,\widehat{V}_S} \left(1 - v_S\right)^{n - n\,\widehat{V}_S} \frac{v_S^{a-1}\left(1 - v_S\right)^{b-1}}{B(a,b)\left(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\right)} \,\mathrm{d}v_S \\
&= \binom{n}{n\,\widehat{V}_S} \frac{B\left(a + n\,\widehat{V}_S,\, b + n - n\,\widehat{V}_S\right)\left(1 - F_{\mathrm{Beta}\left(a + n\,\widehat{V}_S,\, b + n - n\,\widehat{V}_S\right)}\left(\frac{1}{2}\right)\right)}{B(a,b)\left(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\right)},
\end{aligned}$$

where $L\left(\widehat{V}_S \mid H_0\right)$ and $L\left(\widehat{V}_S \mid H_1\right)$ are the marginal likelihoods under the null and alternative hypotheses, respectively, $B(a,b)$ is the Beta function and $F_{\mathrm{Beta}(a,b)}$ is the CDF of the $\mathrm{Beta}(a,b)$ distribution. Then, the Bayes factor in favor of the alternative hypothesis is

$$\begin{aligned}
BF_n &= \frac{L\left(\widehat{V}_S \mid H_1\right)}{L\left(\widehat{V}_S \mid H_0\right)} \\
&= \frac{B\left(a + n\,\widehat{V}_S,\, b + n - n\,\widehat{V}_S\right)\left(1 - F_{\mathrm{Beta}\left(a + n\,\widehat{V}_S,\, b + n - n\,\widehat{V}_S\right)}\left(\frac{1}{2}\right)\right)}{B(a,b)\left(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\right)}\, 2^n.
\end{aligned}$$

Notice that the distribution of BF_n under H_0 has closed form: since $n\,\widehat{V}_S \in \{0, \ldots, n\}$, the Bayes factor takes at most $n+1$ distinct values $x_0, \ldots, x_n$ with

$$\mathbb{P}(BF_n = x_k \mid H_0) = \binom{n}{k} \frac{1}{2^n}, \quad k = 0, \ldots, n,$$

where

$$x_k = \frac{B(a+k,\, b+n-k)\, \left(1 - F_{\mathrm{Beta}(a+k,\, b+n-k)}\left(\frac{1}{2}\right)\right)}{B(a,b)\, \left(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\right)}\, 2^n, \quad k = 0, \ldots, n.$$

Finally, to compute the probability on the left-hand side of Equation (11), we observe that

$$\mathbb{P}\left(BF_n = x_k \mid V_S = v_S > \frac{1}{2}\right) = \binom{n}{k} v_S^k (1 - v_S)^{n-k}, \quad k = 0, \ldots, n.$$

Web Appendix F. Simulation Study: 1Data-Generating Processes and Implementation Details

This appendix describes the data-generating processes and the Markov chain Monte Carlo implementation details for the simulation study in Section 4 of the main paper.

Web Appendix F.1 *Data-Generating Processes*

In all six settings, the treatment indicator $Z_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}\left(\frac{1}{2}\right)$ is drawn independently of all potential outcomes. The data-generating processes for P_i in each setting are as follows.

- **Setting 1 (Useless surrogate):** The potential outcomes are generated as follows:

$$\begin{aligned} Y_{1i} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(5, 3), \quad S_{1i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(\tfrac{2}{3}, 1\right), \\ Y_{0i} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(3, 3), \quad S_{0i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(\tfrac{2}{3}, 1\right), \end{aligned} \qquad i = 1, \ldots, n.$$

  Therefore, by the properties of Gaussian distributions, we have that

$$P_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(\mu^*, \Sigma^*), \quad i = 1, \ldots, n,$$

  where

$$\mu^* = \left(5,\, \tfrac{2}{3},\, 3,\, \tfrac{2}{3}\right)^\top, \qquad \Sigma^* = \mathrm{diag}(3,\, 1,\, 3,\, 1).$$

- **Setting 2 (Perfect surrogate):** The potential outcomes are generated as follows:

$$\begin{aligned} Y_{1i} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(6,3), \quad S_{1i} \mid Y_{1i} \overset{\text{ind.}}{\sim} \mathcal{N}(Y_{1i}, 0.1), \\ Y_{0i} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(2.5,3), \quad S_{0i} \mid Y_{0i} \overset{\text{ind.}}{\sim} \mathcal{N}(Y_{0i}, 0.1), \end{aligned} \qquad i = 1, \ldots, n.$$

Therefore, by the properties of Gaussian distributions, we have that

$$P_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(\mu^*, \Sigma^*)\,, \quad i = 1, \ldots, n,$$

where

$$\mu^* = (6,\, 6,\, 2.5,\, 2.5)^\top, \qquad \Sigma^* = I_2 \otimes \begin{bmatrix} 3 & 3 \\ 3 & 3.1 \end{bmatrix}.$$

where $\otimes$ denotes the Kronecker product.

- **Setting 3 (Imperfect surrogate):** The potential outcomes are generated as follows:

$$\begin{aligned} S_{1i} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(5,3), \quad Y_{1i} \mid S_{1i} \overset{\text{ind.}}{\sim} \mathcal{N}(S_{1i} + 1.5,\, 0.6), \\ S_{0i} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(3,3), \quad Y_{0i} \mid S_{0i} \overset{\text{ind.}}{\sim} \mathcal{N}(S_{0i},\, 0.6), \end{aligned} \qquad i = 1, \ldots, n.$$

Therefore, by the properties of Gaussian distributions, we have that

$$P_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(\mu^*, \Sigma^*)\,, \quad i = 1, \ldots, n,$$

where

$$\mu^* = (6.5,\, 5,\, 3,\, 3)^\top, \qquad \Sigma^* = I_2 \otimes \begin{bmatrix} 3.6 & 3 \\ 3 & 3 \end{bmatrix}.$$

- **Setting 4 (Misspecified model):** The potential outcomes are generated as follows:

$$\begin{aligned} S_{1i} &\overset{\text{i.i.d.}}{\sim} \text{LogNormal}(2.5,\, 1.5^2), \\ Y_{1i} \mid S_{1i} &\overset{\text{ind.}}{\sim} 2.0 + 1.2\sqrt{S_{1i}} + 0.3\, e^{S_{1i}/500} + \text{LogNormal}(0,\, 0.3), \\ S_{0i} &\overset{\text{i.i.d.}}{\sim} \text{LogNormal}(0.5,\, 1.5^2), \\ Y_{0i} \mid S_{0i} &\overset{\text{ind.}}{\sim} 0.8\sqrt{S_{0i}} + 0.2\, e^{S_{0i}/50} + \text{LogNormal}(0,\, 0.3), \end{aligned} \qquad i = 1, \ldots, n.$$

- **Setting 5 (Binary covariate):** The potential outcomes are generated as follows:

$$\begin{aligned} P_i \mid X_i &\overset{\text{ind.}}{\sim} X_i\, \mathcal{N}_4(\mu^1, \Sigma^1) + (1 - X_i)\, \mathcal{N}_4(\mu^0, \Sigma^0)\,, \\ X_i &\overset{\text{i.i.d.}}{\sim} \text{Bernoulli}\left(\tfrac{1}{2}\right), \end{aligned} \qquad i = 1, \ldots, n,$$

where

$$\mu^0 = (5, 5, 0, 0)^\top, \qquad \mu^1 = (5, -5, 0, -10)^\top,$$

$$\Sigma^0 = I_2 \otimes \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \Sigma^1 = I_2 \otimes \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix},$$

- **Setting 6 (Gaussian covariate):** The potential outcomes are generated as follows:

$$\begin{aligned} P_i \mid X_i &\overset{\text{ind.}}{\sim} \mathcal{N}_4(X_i\,\beta,\, \Sigma)\,, \\ X_i &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(m,\, v^2), \end{aligned} \qquad i = 1, \ldots, n,$$

where

$$\beta = (1,\, 7,\, 0,\, 6)^\top, \qquad \Sigma = \tfrac{1}{2}\, I_4,$$

$$m = 3, \qquad v^2 = 1.$$

Web Appendix F.2 *Simulation Parameters*

The surrogate validation threshold η was computed using significance level $\alpha = 0.05$, desired power $1 - \beta = 0.8$, and a uniform prior for V_S with $a = b = 1$.

Each setting was replicated $R = 500$ times. For both BSET implementations, we ran $C = 2$ Markov chains of $T = 500$ iterations each, discarding the first $b = 125$ as burn-in. The prior parameters for each Bayesian model are as follows.

- **Parast et al. (2024):**

$$\mu_0 = (0, 0, 0, 0)^\top, \qquad \Sigma_0 = 10\, I_4, \qquad s = 2, \qquad \tau = 1.$$

- **Carlotti and Parast (2026):**

$$\mu_\beta = (0, 0, 0, 0)^\top, \qquad \Sigma_\beta = 10\, I_4, \qquad s = 2, \qquad \tau = 1.$$

References

Schwartz, L. (1965). On bayes procedures. *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* **4,** 10–26.